

M.Tech. (Online) — Artificial Intelligence
Indian Institute of Science, Bangalore

Literature Review & Benchmark Survey

Detection of AI-Generated and Cloned Speech to Strengthen Voice-Based Authentication in Banking Environments

Student Name	Rajib Roy
Student Number	24459
Programme	M.Tech. in Artificial Intelligence
Internal Guide	Dr. Anil Rahate, Practice Head — AI, Wipro Ltd
IISc Mentor	Prof. Chandra Sekhar Seelamantula, Dept. of Electrical Engineering, IISc
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1. Overview and Scope

This document is **Phase 1 Deliverable 1** of the M.Tech. project. It provides a structured survey of the published literature on synthetic speech detection (commonly framed as “anti-spoofing” in the automatic speaker verification community), with a focus on methods, datasets, evaluation protocols, and robustness findings relevant to the project’s banking telephony deployment context.

Scope of this document: This is a survey of *existing published work*. It does not report results from this project’s own models, dataset, or experiments, as those activities are planned for Phase 2. The purpose is to establish what the published literature says, identify gaps, and record the design decisions taken as a consequence — decisions that will govern the methodology implemented in Phases 2 and 3.

The survey covers five areas:

1. The ASVspoof challenge series — the field’s primary benchmarking platform.
2. Classical and handcrafted feature approaches.
3. Deep learning approaches.
4. Robustness, codec effects, and domain mismatch.
5. Gaps in the existing literature and how this project addresses them.

2. The ASVspoof Challenge Series

2.1 Background

The ASVspoof challenge series, established in 2015 and running through 2021, is the primary benchmarking platform for synthetic speech detection (Nautsch et al., 2021). Each edition defined standardised datasets, evaluation protocols, and baseline systems, enabling reproducible comparison across research groups.

The challenge distinguishes two principal attack scenarios:

- **Logical Access (LA):** Attacks generated by text-to-speech (TTS) synthesis or voice conversion algorithms. This is the attack type most relevant to banking fraud.
- **Physical Access (PA):** Replay attacks, where genuine recordings are replayed through a loudspeaker at a microphone.

2.2 ASVspoof 2019

ASVspoof 2019 (Todisco et al., 2019) is the most directly relevant edition to this project. Its LA dataset covers 19 TTS and voice-conversion systems (A01–A19), providing broad coverage of synthesis architectures from that period.

Key contributions:

- The **Constant-Q Cepstral Coefficient (CQCC)** feature was introduced as the official baseline, achieving $EER \approx 0.10$ with a GMM classifier on the LA track. CQCC provides finer frequency resolution at low frequencies than standard MFCCs, better capturing fine spectral structure in vocoder output.
- The **Equal Error Rate (EER)** — the point on the detection error tradeoff curve where the false acceptance rate equals the false rejection rate — was established as the primary evaluation metric for the field.
- ASVspoof 2019 generated the most widely-used benchmark dataset for subsequent anti-spoofing research, covering statistical parametric, neural, and hybrid TTS systems.

2.3 ASVspoof 2021

ASVspoof 2021 extended the challenge to evaluate systems under **matched and mismatched channel conditions**:

- Systems trained on clean studio recordings degraded substantially under telephone channel evaluation, with EER increases of 5–15 percentage points reported across participating systems.
- The challenge demonstrated that codec compression and channel noise modify both the genuine speech characteristics and the spoofing artefacts the detectors rely upon, making clean-data-trained systems unreliable in real deployments.
- This finding represents a critical gap between published benchmark performance and real-world banking telephony deployment, where audio is transmitted over G.711 or similar codecs at 8 kHz.

2.4 Detection Cost Function (t-DCF)

Kinnunen et al. (2020) introduced the tandem Detection Cost Function (t-DCF) as a more operationally realistic performance measure:

- Unlike EER, t-DCF jointly assesses the combined error rate of a spoofing countermeasure and an ASV system, accounting for the prior probability of attack and the relative cost of false acceptance versus false rejection.
- This reflects the fact that a countermeasure is not used in isolation — it operates as a front-end to an existing speaker verification system.
- t-DCF has been adopted as a co-primary metric alongside EER in ASVspoof 2021.

3. Classical and Handcrafted Feature Approaches

3.1 Systematic Feature Comparison

Sahidullah et al. (2015) provided the most comprehensive comparison of acoustic features for anti-spoofing at the time of ASVspoof 2015:

- Features capturing **fine spectral structure** consistently outperformed standard 13-coefficient MFCCs against vocoder-based attacks. Specifically, using 40 MFCC coefficients (capturing higher-order cepstral details) outperformed the conventional 13-coefficient setting by a substantial margin.
- The inclusion of **delta and delta-delta coefficients** (encoding temporal rate-of-change of the spectral envelope) provided consistent gains across all tested classifiers. This is attributed to the difference in spectral dynamics between genuine articulation — which produces continuously evolving spectral envelopes — and TTS vocoders, which impose smoother and more regular temporal evolution.
- **Harmonic-to-Noise Ratio (HNR)** and **spectral shape features** (centroid, bandwidth, rolloff) were identified as informative secondary features, reflecting the characteristically clean harmonic structure of TTS vocoders.

3.2 Phase Spectrum Features

Patel and Patil (2015) demonstrated that TTS vocoders introduce **phase coherence artefacts** that are invisible in the magnitude spectrum:

- Vocoders generate speech with artificially coherent phase structures, whereas natural speech exhibits complex and partially random phase relationships across frames.
- Features derived from the instantaneous phase spectrum, group delay, or phase difference between adjacent frames were shown to capture vocoder artefacts that MFCC-based features miss entirely.
- Combining phase-based features with MFCC features yielded additive accuracy improvements on ASVspoof 2015, motivating their inclusion in broader feature sets.

3.3 SVM Classifiers for Anti-Spoofing

Support Vector Machines with RBF kernels have been the dominant classifiers for hand-crafted features throughout the ASVspoof 2015–2019 era:

- SVMs generalise well with relatively small training sets, compared to the large datasets needed by deep learning models, and have strong theoretical foundations for binary classification in high-dimensional feature spaces.
- The RBF kernel can model non-linear decision boundaries without requiring explicit feature interaction terms.
- Probabilistic calibration via Platt scaling enables threshold tuning at the EER operating point, which is important for class-imbalanced detection problems.
- The official ASVspoof 2019 baseline used SVM with CQCC features, and SVM remains a strong and reproducible reference system in the field.

3.4 Summary of Findings

Key Takeaways — Classical Features

- Delta and delta-delta MFCCs are the strongest discriminators: they capture temporal rate-of-change, which is smoother and more regular in TTS output.
- 40 MFCC coefficients substantially outperform the conventional 13-coefficient setting for vocoder detection.
- Phase coherence features capture artefacts invisible in the magnitude spectrum, providing complementary and non-redundant discriminative information.
- SVM with RBF kernel is a well-validated, reproducible classical baseline.

4. Deep Learning Approaches

4.1 CNN-Based Systems

The transition to deep learning in anti-spoofing accelerated with ASVspoof 2019. Alzantot et al. (2019) showed that a ResNet architecture trained end-to-end on log-Mel spectrograms could match the CQCC + SVM official baseline:

- The ResNet learns hierarchical spectral representations directly from the input spectrogram, eliminating hand-engineered features.
- The model used **global average pooling across the time dimension** to produce a fixed-length representation from variable-length utterances.
- This temporal pooling operation discards the order and dynamics of the spectral sequence, retaining only average spectral content. This is a known limitation when the discriminative signal includes temporal patterns — which vocoder artefacts do.

4.2 Hybrid CNN-LSTM Architectures

Multiple subsequent studies have found that hybrid CNN-LSTM architectures consistently outperform purely convolutional models for anti-spoofing:

- The **CNN front-end** acts as a local spectral feature extractor, identifying frequency-domain artefacts within short analysis windows.
- The **LSTM layer** then models the temporal evolution of these spectral features across the utterance, capturing prosodic regularity and temporal dynamics that CNN global pooling discards.
- Using **frequency-only pooling** (pooling across frequency but not time) after the CNN preserves the temporal sequence, enabling the LSTM to process the full temporal context.
- Bidirectional LSTMs, which process the sequence in both forward and backward directions, capture dependencies that span the full utterance and have shown further improvements.

4.3 Self-Supervised Representations: wav2vec 2.0

Baevski et al. (2020) introduced wav2vec 2.0, a transformer-based model pre-trained on 960 hours of unlabelled speech via a contrastive self-supervised objective:

- When the frozen or fine-tuned wav2vec 2.0 encoder is used for spoofing detection, it achieves competitive EERs without any task-specific feature engineering, suggesting that self-supervised pre-training learns representations that are also discriminative for synthetic speech detection.
- Full fine-tuning requires substantial GPU memory (16 GB+) and training time. Feature extraction from a frozen encoder is computationally tractable and still competitive.
- This line of work suggests that large pre-trained speech models encode properties of natural speech that synthetic systems fail to fully replicate.

4.4 Graph Attention: AASIST

Wang et al. (2021) introduced AASIST (Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks):

- AASIST applies graph attention networks to model relationships between nodes spanning both spectral and temporal dimensions simultaneously, capturing cross-frequency and cross-time dependencies not modelled by CNNs or LSTMs independently.
- It achieves $\text{EER} \approx 0.02$ on ASVspoof 2021 LA — the state of the art at the time of publication.
- With approximately 300,000 parameters, AASIST is relatively compact for its performance level, making it practically deployable.

4.5 Comparative Published Performance

Table 1 summarises indicative EER values from key published anti-spoofing systems on ASVspoof datasets. These numbers represent the state of the art surveyed during Phase 1 and provide context for understanding the field's progress.

Reference	Dataset	Method	EER ↓	Notes
Todisco et al. (2019)	ASVspoof 2019 LA	CQCC + GMM	~0.10	Official baseline
Alzantot et al. (2019)	ASVspoof 2019 LA	ResNet spectrogram	~0.05	CNN-based
Wang et al. (2021)	ASVspoof 2021 LA	AASIST	~0.02	SOTA at publication
Tak et al. (2022)	ASVspoof 2021 LA	RawBoost + ResNet	~0.015	Data augmentation

Table 1: Indicative EER of key published anti-spoofing systems. Values are approximate; exact results vary with evaluation protocol. This table represents the published state of the art at the time of the Phase 1 literature survey.

4.6 Summary of Findings

Key Takeaways — Deep Learning

- CNN-based systems match classical CQCC baselines but are limited by global temporal pooling, which discards prosodic dynamics.
- CNN-LSTM hybrids with frequency-only pooling consistently outperform CNNs alone by modelling temporal evolution of spectral features.
- Self-supervised models (wav2vec 2.0) and graph attention models (AASIST) represent the current frontier, achieving EER below 0.02 on ASVspoof 2021.
- The field has progressed rapidly from EER ≈ 0.10 (CQCC+GMM, 2019) to EER ≈ 0.015 (RawBoost+ResNet, 2022) on standardised benchmarks.

5. Robustness, Codec Effects, and Domain Mismatch

5.1 Channel Mismatch and Codec Compression

A critical finding across the ASVspoof challenge series is that systems trained on clean benchmark data degrade significantly under realistic channel conditions:

- Yi et al. (2022), through the ADD 2022 challenge, explicitly evaluated systems under telephone channel conditions, finding EER increases of 5–15 percentage points compared to clean evaluation.
- G.711 μ -law codec compression (8 kHz, used in PSTN and banking IVR systems) introduces quantisation noise and spectral smoothing that modifies both genuine speech characteristics and the synthetic artefacts that detectors rely upon.
- ADD 2022 found that codec compression causes moderate and predictable degradation, while additive noise (particularly at low SNR) causes more severe and less predictable degradation. The consensus finding is that **noise is a more serious deployment threat than codec compression** for anti-spoofing systems in telephony environments.

5.2 Cross-System Generalisation

Kinnunen et al. (2020) investigated how well systems trained on ASVspoof 2019 TTS systems generalise to newer or unseen TTS architectures:

- Systems achieving near-zero EER on ASVspoof 2019 test systems sometimes failed on newer TTS architectures that were not present in the training data.
- This generalisation failure arises when models learn **system-specific artefacts** (peculiar to particular TTS implementations) rather than **general properties** common

to all neural TTS vocoders.

- Models trained on **diverse** sets of TTS systems, covering multiple synthesis paradigms (statistical parametric, attention-based, flow-based, GAN-based), tend to generalise better to unseen systems than those trained on a narrow set.

5.3 Data Augmentation Strategies

Tak et al. (2022) demonstrated that **RawBoost** — a signal-level data augmentation technique applied during training — significantly improves robustness:

- RawBoost applies random combinations of coloured noise, impulsive noise, and stationary background noise to training waveforms, forcing the model to learn features that are stable under noise variation.
- This training-time augmentation approach outperformed the application of noise reduction at inference time, showing that robustness must be built into the model rather than compensated for at deployment.
- RawBoost combined with ResNet achieved EER ≈ 0.015 on ASVspoof 2021 LA — the best published result at the time of this survey.

5.4 Summary of Findings

Key Takeaways — Robustness and Domain Mismatch

- Training on clean data and hoping for channel generalisation does not work — EER increases of 5–15 percentage points are reported under telephone conditions.
- Noise is a more severe deployment threat than codec compression alone. Both must be addressed, but noise deserves priority attention.
- Cross-system generalisation is a known failure mode. Training on a diverse set of TTS systems covering multiple synthesis paradigms improves robustness to unseen systems.
- Training-time noise augmentation is more effective than inference-time noise reduction for building robust detectors.

6. Gaps in the Literature and Project Positioning

6.1 Identified Gaps

Despite the maturity of the ASVspoof benchmark and a large body of published systems, three gaps in the existing literature are directly relevant to this project:

1. **Banking telephony simulation:** The overwhelming majority of published work evaluates on clean studio-quality recordings, or on the specific ASVspoof 2021 evaluation channel. There is comparatively little published work that explicitly simulates banking IVR telephony conditions: 8 kHz sampling rate, G.711 μ -law codec, and short transactional utterances (3–8 seconds of banking phrases rather than long read sen-

tences).

2. **Controlled head-to-head comparison:** Most published work focuses exclusively on either classical feature-based systems or deep learning systems. Few studies directly compare both under *identical data and conditions*, making it difficult to quantify the practical benefit of deep learning over well-engineered handcrafted features in a specific deployment context.
3. **Robustness degradation curves:** Published results typically report a single performance number at a specific evaluation condition. Reporting *continuous degradation curves* as a function of SNR level and codec type is rare in the anti-spoofing literature, yet this information is essential for deployment engineering decisions (e.g., setting operating thresholds under expected noise conditions).

6.2 Design Decisions Motivated by the Literature

Table 2 maps each design decision in this project’s methodology to the specific literature finding that motivated it. These decisions govern the dataset construction, feature engineering, and model development planned for Phases 2 and 3.

Design Decision	Literature Motivation	How It Will Be Implemented
40 MFCC coefficients (not 13)	Sahidullah et al. (2015): fine spectral structure discriminates vocoders	<code>n_mfcc=40</code> in <code>config.yaml</code> ; 257-dim vector
Delta + delta-delta MFCCs	Sahidullah et al. (2015): temporal dynamics are the primary discriminative signal	120 of 257 dimensions encode temporal rate-of-change
Phase difference features	Patel & Patil (2015): phase coherence artefacts invisible in magnitude spectrum	2 dimensions; supplements MFCC information
8 kHz + G.711 preprocessing	ASVspoof 2021; Yi et al. (2022): channel mismatch degrades clean-trained models	All utterances processed through codec simulation pipeline
EER as primary metric	ASVspoof series standard; balances FAR and FRR at single threshold	Threshold set at EER point on validation set
CNN-LSTM as planned DL architecture	Multiple studies: temporal modelling via LSTM improves over CNN-only	2-layer BiLSTM on time-preserving spectrogram front-end
Edge-TTS as held-out test	Kinnunen et al. (2020): cross-system generalisation is a known failure mode	1,133 clips from 20 commercial voices; zero training overlap
5/10/15/20 dB noise variants	Yi et al. (2022); Tak et al. (2022): noise is primary deployment threat	Dataset pipeline produces augmented variants; planned for Phase 2 training
Diverse TTS training set	Kinnunen et al. (2020): diverse training improves generalisation	ASVspoof 2019 (6 systems) + VITS + Tacotron2
SVM as classical baseline	ASVspoof 2019 official baseline; well-validated in the literature	RBF kernel with EER-optimal threshold on validation set

Table 2: Literature-to-implementation design decision mapping. These decisions govern the methodology for Phases 2 and 3.

Note on scope: The decisions in Table 2 describe *planned* methodology for Phase 2, informed by the Phase 1 literature survey. They are included here to make the traceability between literature findings and project design explicit. The actual implementation and results will be reported in the Phase 2 interim report.

7. Summary

This literature review has surveyed the field of synthetic speech detection from the foundational ASVspoof 2015–2021 challenges through to the current state of the art. The principal conclusions for this project are:

1. **Feature design:** Delta-MFCC coefficients (40 coefficients, with Δ and $\Delta\Delta$) are the strongest discriminators between genuine and synthetic speech. Phase-based features provide complementary, non-redundant information. Both will be included in the planned handcrafted feature vector.
2. **Classifier choice:** SVM with RBF kernel is the appropriate classical baseline, consistent with ASVspoof field convention. CNN-LSTM hybrids are the preferred deep learning architecture, motivated by consistent superiority over CNN-only systems in the temporal modelling of spoofing artefacts.
3. **Telephony simulation:** Training under realistic banking channel conditions (8 kHz, G.711) from the outset is essential, not optional. The literature clearly shows that clean-trained models fail under channel mismatch.
4. **Robustness evaluation:** Noise is the primary deployment threat. Degradation curves across SNR levels (5–20 dB) will be produced in Phase 2 to quantify this empirically for the five planned models.
5. **Generalisation test:** A held-out system (Microsoft Edge-TTS) not present in any published anti-spoofing training set will be used to measure cross-system generalisation directly, addressing the gap identified by Kinnunen et al. (2020).

References

- Alzantot, M., Wang, Z., and Srivastava, M.B. (2019). Deep Residual Neural Networks for Audio Spoofing Detection. *Proc. Interspeech 2019*, 1078–1082.
- Baevski, A., Zhou, Y., Mohamed, A., and Auli, M. (2020). wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. *Advances in Neural Information Processing Systems (NeurIPS)*, 33.
- Kim, J., Kong, J., and Son, J. (2021). Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech. *ICML 2021*.
- Kinnunen, T., Sahidullah, M., Falcone, M., et al. (2020). t-DCF: a Detection Cost Function for the Tandem Assessment of Spoofing Countermeasures and Automatic Speaker Verification. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*.
- McFee, B., Raffel, C., Liang, D., et al. (2015). librosa: Audio and Music Signal Analysis in Python. *Proceedings of the 14th Python in Science Conference*, 18–25.
- Nautsch, A., Wang, X., Evans, N., et al. (2021). ASVspoof 2019: Spoofing Countermeasures for the Detection of Synthesized, Converted and Replayed Speech. *IEEE*

Transactions on Biometrics, Behavior, and Identity Science.

- Panayotov, V., Chen, G., Povey, D., and Khudanpur, S. (2015). Librispeech: An ASR Corpus Based on Public Domain Audio Books. *ICASSP 2015*.
- Patel, T.B. and Patil, H.A. (2015). Combining Evidences from Mel Cepstral, Cochlear Filter Cepstral and Instantaneous Frequency Features for Detection of Natural vs. Spoofed Speech. *Interspeech 2015*, 2062–2066.
- Sahidullah, M., Kinnunen, T., and Hanilci, C. (2015). A Comparison of Features for Synthetic Speech Detection. *Interspeech 2015*, 2087–2091.
- Tak, H., Patino, J., Todisco, M., et al. (2022). End-to-End Spectro-Temporal Graph Attention Networks for Speaker Verification Anti-Spoofing and Speech Deepfake Detection. *ASVspoof 2021 Workshop*.
- Todisco, M., Wang, X., Vestman, V., et al. (2019). ASVspoof 2019: Future Horizons in Spoofed and Fake Speech Detection. *Interspeech 2019*.
- van den Oord, A., Dieleman, S., Zen, H., et al. (2016). WaveNet: A Generative Model for Raw Audio. *arXiv:1609.03499*.
- Wang, T., Yi, J., Fu, R., et al. (2021). AASIST: Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks. *ICASSP 2022*.
- Wang, Y., Skerry-Ryan, R.J., Stanton, D., et al. (2017). Tacotron: Towards End-to-End Speech Synthesis. *Interspeech 2017*.
- Yi, J., Fu, R., Tao, J., et al. (2022). ADD 2022: The First Audio Deep Synthesis Detection Challenge. *ICASSP 2022*.